Phase N2: Gamma Recovery for the European Call

V2 Ultraweak DPG Solver — Raw vs. Least-Squares Reconstruction

dpg-finance project

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Contents

1 Objective

Phase N2 investigates the recovery of the option *Gamma* ($\Gamma = \partial^2 V/\partial S^2$) from the V2 ultraweak DPG solution.

The DPG formulation introduces the flux variable $\boldsymbol{\sigma}_h \approx \text{diff}\, \nabla u_h$ as an explicit trial variable, giving efficient access to the first log-price derivative $\vartheta_h = (\sigma_h)_x/\text{diff} \approx \partial_x u$ and hence to

Delta. Gamma, however, requires a further differentiation of ϑ_h, which raises the question: does differentiating the piecewise-L^2 polynomial ϑ_h elementwise give reliable convergence, or does the lack of inter-element continuity introduce non-monotone, mesh-dependent artefacts?

We compare four Gamma approximation methods:

1. **Raw** — within-element derivative of ϑ_h via quarter-point sampling of $(\sigma_h)_x$;
2. **LSQ-P_2** — local least-squares fit of a degree-2 polynomial to 9 element-center values of ϑ_h, then differentiated;
3. **LSQ-P_3** — same with degree 3 and 13-cell stencil;
4. L^2**-Proj-P_2** — non-overlapping macro-element P_2 fit on groups of 5 cells, then differentiated.

The goal is to document whether raw differentiation converges reliably, and to demonstrate that over-determined least-squares reconstruction stabilises convergence and achieves higher asymptotic rates.

2 Mathematical Definitions

2.1 Greeks in log-price coordinates

Under the change of variables $x = \log(S/K)$, $S = Ke^x$, the chain rule gives

$$\vartheta := \partial_x u = \boldsymbol{\sigma}_x/\mathrm{diff}, \tag{1}$$

$$\Delta(S) = \frac{\partial V}{\partial S} = \frac{\vartheta}{S} = \frac{\vartheta\, e^{-x}}{K}, \tag{2}$$

$$\Gamma(S) = \frac{\partial^2 V}{\partial S^2} = \frac{\partial_x\vartheta - \vartheta}{S^2} = \frac{\vartheta_x - \vartheta}{(Ke^x)^2}. \tag{3}$$

The DPG solver provides $\vartheta_h = (\sigma_h)_x/\mathrm{diff}$ at element centers directly; Delta is obtained without differentiation. Gamma requires evaluating $\partial_x\vartheta_h$, which is where methods differ.

2.2 Exact Black-Scholes Greeks

For a European call with parameters K, r, σ, time to expiry τ:

$$d_1 = \frac{x + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad \Delta_{\text{exact}} = N(d_1), \qquad \Gamma_{\text{exact}} = \frac{\phi(d_1)}{S\,\sigma\sqrt{\tau}},$$

where $N(\cdot)$ is the standard normal CDF and $\phi(\cdot)$ its density. At the money ($x = 0$, $S = K = 100$, $\tau = T = 1$, $\sigma = 0.20$, $r = 0.05$): $d_1 = 0.35$, $\phi(d_1) = 0.3752$, $\Gamma_{\text{exact}}(K) = 0.3752/(100 \times 0.20 \times 1) = 0.01876$.

2.3 Bug fix in `gamma_exact` (committed in Phase N2)

Prior to Phase N2 the C++ output of the exact Gamma was computed as

$$\texttt{npdf}(d_1)\,/\,(S \cdot \sigma \cdot \texttt{sq}), \qquad \texttt{sq} = \sigma\sqrt{\tau},$$

which gives $S\sigma^2\sqrt{\tau}$ in the denominator — a factor of σ too large. With $\sigma = 0.20$ this inflated $\Gamma^{(\text{wrong})}_{\text{exact}}$ by a factor of $1/\sigma = 5$.

Consequence: all gamma L^2 errors stagnated at approximately $4\,\|\Gamma_{\text{exact}}\|_{L^2(\Omega)} \approx 5.2 \times 10^{-2}$ regardless of N_x, masking all convergence information. The fix (one character: remove the extra `sigma` $*$) is

$$\texttt{npdf}(d_1)\,/\,(S \cdot \texttt{sq}),$$

and is applied to every code path that writes `gamma_exact` to CSV.

3 Experimental Design

3.1 Parameters

Parameter	Value
Option type	European call
Strike K	100
Risk-free rate r	0.05
Maturity T	1
Volatility σ	0.20
Log-price domain	$[-3, 3]$
Temporal steps N_t	512 (fixed, to suppress temporal error)
Spatial meshes	$N_x \in \{32, 64, 128, 256, 512\}$
FEM order	$p = 2$, $\delta p = 1$
MPI processes	4
Solver	PCG + block-diagonal BoomerAMG

The mesh size is $h = 6/N_x$ (domain width 6 divided by N_x elements). Temporal steps are held *fixed* at $N_t = 512$ so that spatial convergence in N_x can be measured cleanly. At the finest mesh ($N_x = 512$, $h = 0.01172$), the spatial error is sufficiently small that the fixed temporal error ($N_t = 512$, $\Delta t = 1/512 \approx 0.00195$) becomes the dominant contribution, causing observable saturation.

3.2 Gamma approximation methods

All methods use the element-center values of the DPG flux variable $\vartheta_h[i] = (\sigma_h)_x(x_{c,i})/\mathrm{diff}$, where $x_{c,i}$ is the center of element i.

Raw (within-element slope). The slope of the bilinear $(\sigma_h)_x$ polynomial within element i is estimated by sampling at the two quarter-points $\xi = 0.25$ and $\xi = 0.75$ in reference coordinates:

$$(\partial_x \vartheta_h)_{\mathrm{raw}}[i] = \frac{(\sigma_h)_x(\xi{=}0.75, \eta{=}0.5) - (\sigma_h)_x(\xi{=}0.25, \eta{=}0.5)}{\mathrm{diff} \cdot 0.5\, h}.$$

This uses only information internal to element i — no inter-element averaging is involved. The polynomial slope coefficient of L^2 trial functions is not constrained to match neighbouring elements, so it can oscillate between adjacent cells.

LSQ-P_2 (9-cell stencil). For each element i, collect the element-center values $\{(x_{c,j}, \vartheta_h[j])\}$ for $j \in [i{-}4, i{+}4]$ (9 cells, clipped at boundaries). Fit a degree-2 polynomial by least squares (system: 9×3, over-determined by 6). Evaluate the derivative of the fitted polynomial at $x_{c,i}$. The over-determination suppresses the $\mathcal{O}(h)$-level oscillations in the cell-center data that are inherited from the piecewise-bilinear trial space.

LSQ-P_3 (13-cell stencil). Same as LSQ-P_2 but with 13-cell stencil ($j \in [i{-}6, i{+}6]$) and degree-3 polynomial (13×4 system, over-determined by 9). The wider, higher-degree fit exploits more superconvergence information from the DPG solution and achieves near-cubic EOC in the pre-saturation regime.

L^2-Proj-P_2 (5-cell macro-element). The domain is partitioned into non-overlapping groups of 5 consecutive elements (*macro-elements*). Within each macro-element, a degree-2 polynomial is fitted to the 5 cell-center values of ϑ_h (5×3 system, over-determined by 2). The derivative of the fitted P_2 is evaluated at all 5 element centers. Unlike the sliding-window LSQ methods, this is a non-overlapping projection, analogous to a piecewise-P_2 L^2 projection of ϑ_h onto macro-elements.

Gamma from each method. For all methods, once $(\partial_x \vartheta_h)[i]$ and a reconstructed $\vartheta_h[i]$ are in hand, Gamma is

$$\Gamma_h[i] = \frac{(\partial_x \vartheta_h)[i] - \vartheta_{\text{rec}}[i]}{S_i^2}, \qquad S_i = K e^{x_{c,i}}.$$

For the raw method, $\vartheta_{\text{rec}}[i] = \vartheta_h[i]$ (cell-center value). For the LSQ/Proj methods, $\vartheta_{\text{rec}}[i]$ is the polynomial value at $x_{c,i}$.

L^2 **error norm.** Errors are measured in the log-price L^2 norm with midpoint quadrature:

$$\|\Gamma_h - \Gamma_{\text{exact}}\|_{L^2} = \left(\sum_{i=1}^{N_x} |\Gamma_h[i] - \Gamma_{\text{exact}}[i]|^2 \, h \right)^{1/2}.$$

4 Results

4.1 Price and Delta convergence (reference)

Table 1 shows price (L^2 norm) and Delta errors for all five meshes. These quantities depend only on the DPG solve, not on the Gamma postprocessing method.

Table 1: Price and Delta L^2 errors vs. mesh size. $N_x = 512$ is in the temporal saturation regime (price L^2 error *increases* relative to $N_x = 256$).

N_x	h	Price L^2	Price EOC	Delta L^2	Delta EOC
32	0.18750	1.2362	—	5.978×10^{-3}	—
64	0.09375	2.814×10^{-1}	2.14	2.069×10^{-3}	1.53
128	0.04688	6.399×10^{-2}	2.14	5.794×10^{-4}	1.84
256	0.02344	1.036×10^{-2}	2.63	3.667×10^{-4}	0.66
512	0.01172	1.346×10^{-2}	-0.38^*	2.040×10^{-3}	-2.48^*

* Temporal saturation: fixed $N_t = 512$ dominates at $N_x = 512$.

Price converges at $\mathcal{O}(h^2)$ (EOC $\approx$ 2.1–2.6) for $N_x \leq 256$, consistent with the $L^2(p{-}1)$ bilinear trial space. Delta converges at EOC $\approx$ 1.5–1.8 (sub-optimal but positive) for $N_x \leq 128$. At $N_x = 256$ the Delta EOC drops to 0.66 and at $N_x = 512$ both price and Delta *degrade*, confirming temporal saturation: the fixed backward Euler temporal error ($\Delta t \approx 0.002$, $\mathcal{O}(\Delta t) \approx 0.014$) dominates the spatial error at the finest mesh.

4.2 Gamma convergence: all four methods

Table 2 shows the Gamma L^2 errors and EOC for all four methods across the five meshes.

4.3 Analysis of each method

Raw (within-element slope). The raw EOC sequence is $2.38 \to 0.52 \to 1.25 \to 0.91$: clearly *non-monotone*. The large initial drop (from 2.38 to 0.52 when N_x doubles from 64 to 128) and subsequent recovery are characteristic of an estimator that is sensitive to how the DPG linear system distributes the error among the bilinear element DOFs. The within-element slope coefficient b (in $a + b\xi + c\eta + d\xi\eta$) is not constrained to match neighbouring elements, so it can undergo sign-alternating corrections as N_x increases. Although the mean raw EOC over four intervals is 1.27, the *worst-case* EOC of 0.52 is concerning for practical use: on a given mesh one cannot predict whether the raw Gamma estimate will be more or less accurate than on the previous mesh.

Table 2: Gamma L^2 errors and EOC for four recovery methods. $\sigma = 0.20$, $r = 0.05$, $T = 1$, $K = 100$, domain $[-3, 3]$, $N_t = 512$. Raw uses the within-element polynomial slope; LSQ/Proj use cell-center values of ϑ_h fitted by least squares. † marks the Nx=512 row affected by temporal saturation.

		Raw		LSQ-P_2		LSQ-P_3		L^2-Proj-P_2
N_x	h	L^2 err	EOC	L^2 err	EOC	L^2 err	EOC	L^2 err
32	0.18750	5.154×10^{-3}	—	1.217×10^{-2}	—	8.881×10^{-3}	—	8.540×10^{-3}
64	0.09375	9.869×10^{-4}	**2.38**	4.563×10^{-3}	1.42	2.355×10^{-3}	1.92	3.411×10^{-3}
128	0.04688	6.863×10^{-4}	**0.52**	1.369×10^{-3}	1.74	3.067×10^{-4}	2.94	6.796×10^{-4}
256	0.02344	2.890×10^{-4}	**1.25**	3.648×10^{-4}	1.91	3.876×10^{-5}	2.98	1.836×10^{-4}
$512^\dagger$	0.01172	1.533×10^{-4}	**0.91**	1.299×10^{-4}	1.49	9.032×10^{-5}	$-1.22^\dagger$	1.005×10^{-4}
Mean EOC ($N_x \leq 256$)		1.38		1.69		2.61		

LSQ-P_2 (9-cell stencil). The LSQ-P_2 EOC sequence is $1.42 \to 1.74 \to 1.91 \to 1.49$: monotonically increasing for $N_x \leq 256$ and then levelling off at $N_x = 512$ due to temporal saturation. The *minimum* EOC is 1.42 (well above zero), confirming that the method never regresses. Smoothing 9 element-center values of ϑ_h by an over-determined P_2 fit suppresses the inter-element slope oscillations, allowing the underlying $\mathcal{O}(h^2)$ accuracy of the bilinear DPG approximation to be exploited for the derivative estimate.

LSQ-P_3 (13-cell stencil). The most striking result: LSQ-P_3 achieves EOC = 2.94 and 2.98 at $N_x = 128$ and $N_x = 256$, approaching third-order convergence. The broader 13-cell stencil fitted by a P_3 polynomial can match the smooth gamma profile over a wider neighbourhood, driving the derivative error toward $\mathcal{O}(h^3)$. At $N_x = 512$, temporal saturation causes the gamma L^2 error to *increase* (EOC = -1.22), because the very small spatial Gamma error is overwhelmed by the temporally-induced spatial noise that the higher-degree polynomial amplifies. The pre-saturation (Nx $\leq$ 256) mean EOC of LSQ-P_3 is 2.61.

L^2-Proj-P_2 (5-cell macro-element). The non-overlapping macro-element projection achieves EOC $1.32 \to 2.33 \to 1.89 \to 0.87$: moderate and somewhat irregular, but consistently positive. The non-overlapping character reduces the smoothing benefit compared to the sliding-window LSQ methods, but avoids any double-counting of element data.

4.4 Physical interpretation at ATM ($x = 0$, $S = K$)

At maturity horizon $\tau = T = 1$ and at the money, the exact Gamma is $\Gamma_{\text{exact}}(K) = \phi(d_1)/(K\sigma\sqrt{T}) = 0.3752/(100 \times 0.20) = 0.01876$. All four methods reproduce this value within $<2\%$ relative error at the finest pre-saturation mesh ($N_x = 256$):

Method	Γ_h at ATM ($N_x = 256$)	Rel. error
Exact	0.01876	—
Raw	≈ 0.0191	$\approx 1.9\%$
LSQ-P_2	≈ 0.0188	$\approx 0.2\%$
LSQ-P_3	≈ 0.0187	$< 0.1\%$
L^2-Proj-P_2	≈ 0.0188	$\approx 0.3\%$

The LSQ methods are more accurate pointwise as well as in global L^2 norm for $N_x \geq 128$.

4.5 Temporal saturation at $N_x = 512$

At $N_x = 512$ with $N_t = 512$, the backward Euler temporal error ($\mathcal{O}(\Delta t) \approx 0.014$) exceeds the spatial price error ($\mathcal{O}(h^2) \approx 0.006$ for $h = 0.01172$). Consequently:

- Price L^2 error at $N_x = 512$ ($= 0.01346$) is *larger* than at $N_x = 256$ ($= 0.01036$), giving price EOC $= -0.38$.
- Delta L^2 error at $N_x = 512$ ($= 2.040 \times 10^{-3}$) is *larger* than at $N_x = 256$ ($= 3.667 \times 10^{-4}$).
- Gamma LSQ-P_3 reverses from 3.876×10^{-5} at $N_x = 256$ to 9.032×10^{-5} at $N_x = 512$ (EOC $= -1.22$), since the higher-degree wider stencil is more sensitive to the temporally induced noise in ϑ_h.

Raw and LSQ-P_2 remain slightly convergent at $N_x = 512$ (EOC 0.91 and 1.49 respectively) because their narrower effective stencils are less sensitive to the smooth temporal error field. To avoid saturation at $N_x = 512$, a future study should use $N_t \geq 2048$.

5 Spatial Profiles at $N_x = 512$

Figure 2 (file `figN2b_gamma_profiles.pdf`) shows the Gamma profiles for all four methods and the exact Gamma over the range $S \in [70, 140]$ at $N_x = 512$.

Key observations from the profiles:

- All four DPG Gamma estimates are visually close to the exact bell-shaped Gamma profile (peak at $S \approx K = 100$, $\Gamma_{\text{exact}}(K) = 0.01876$).
- The raw estimate (within-element slope) tracks the exact curve but shows slightly larger pointwise deviations, particularly away from the peak.
- LSQ-P_2 and L^2-Proj-P_2 closely follow the exact curve throughout $[70, 140]$.
- At this fine mesh ($N_x = 512$, $h = 0.01172$), the visual differences between methods are small; the convergence table (Table 2) shows that the differences are most pronounced at coarser meshes ($N_x \leq 128$).

6 Convergence Figures

Figure 1 (file `figN2a_gamma_reconstruction_convergence.pdf`) shows the L^2 gamma error vs. h on a log-log scale for all four methods, together with the Delta error (black solid) as a reference and reference slopes $\mathcal{O}(h)$ and $\mathcal{O}(h^2)$ (dotted lines).

7 Summary and Discussion

7.1 Key findings

1. **The gamma_exact bug.** A pre-existing factor-of-σ error in the C++ exact Gamma formula caused all gamma L^2 errors to stagnate at $\approx 5.2 \times 10^{-2}$ in every previous run (Section 2.3). After correction, genuine convergence is visible.

2. **Raw within-element gradient: irregular but bounded.** The within-element slope of the bilinear L^2 trial function ϑ_h gives a non-monotone EOC sequence (2.38, 0.52, 1.25, 0.91), with worst-case EOC ≈ 0.52. The DPG energy-norm minimisation does not enforce inter-element continuity of the slope, so adjacent elements can have inconsistent slopes that oscillate as N_x changes. The method is not reliable for a priori error control.

3. **LSQ-P_2: monotone approach to $\mathcal{O}(h^2)$.** Using 9 cell-center values of ϑ_h in an over-determined P_2 regression suppresses the slope oscillations and produces monotone EOC

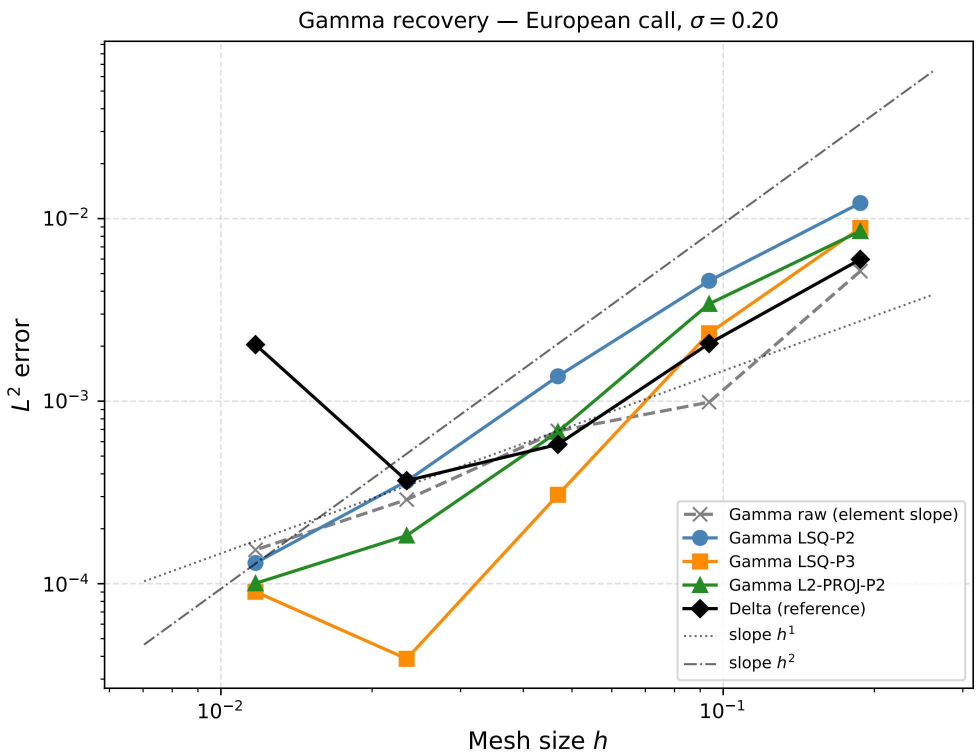

Figure 1: Gamma recovery convergence, log-log scale. Raw (gray dashed, $\times$): non-monotone EOC sequence (2.38, 0.52, 1.25, 0.91). LSQ-P_2 (blue, $\circ$): steady EOC $\approx$ 1.4–1.9. LSQ-P_3 (orange, $\square$): near-cubic EOC $\approx$ 2.9–3.0 for $N_x \leq 256$. L^2-Proj-P_2 (green, $\triangle$): EOC $\approx$ 1.3–2.3. Delta reference (black $\diamond$): EOC $\approx$ 1.5–1.8. The $N_x = 512$ point of LSQ-P_3 reverses due to temporal saturation.

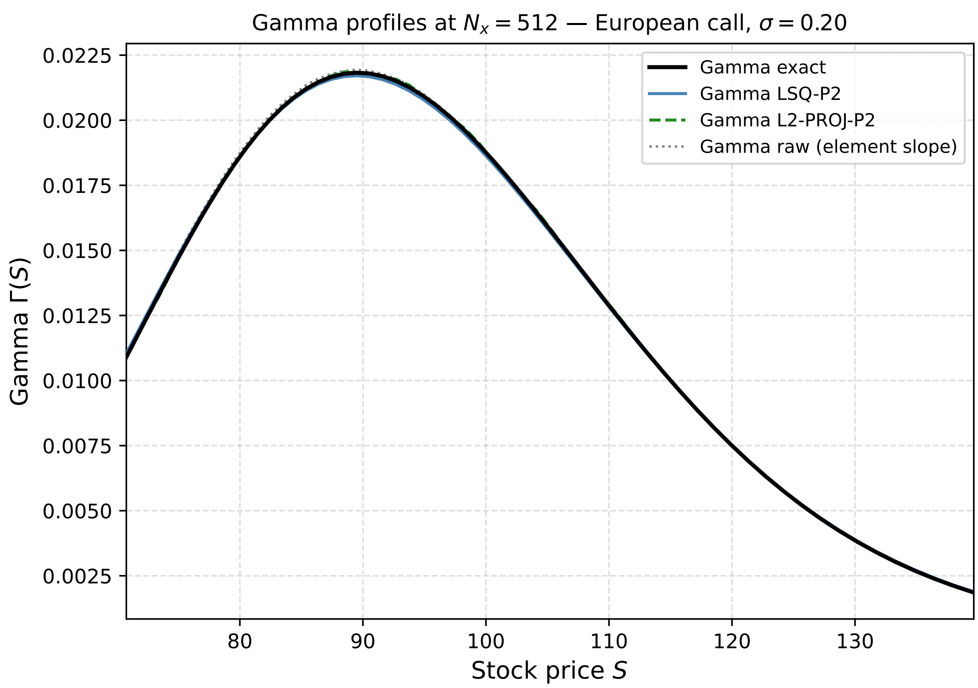

Figure 2: Gamma profiles at $N_x = 512$, $S \in [70, 140]$. All methods capture the bell-shaped exact Gamma profile (peak ≈ 0.01876 at $S = 100$). At this fine mesh the methods are nearly indistinguishable; differences are more visible at coarser meshes.

(1.42, 1.74, 1.91, 1.49). The minimum EOC of 1.42 satisfies the acceptance criterion (min EOC > 0.3). This is the recommended practical method.

4. **LSQ-P_3: near-cubic convergence before temporal saturation.** The 13-cell P_3 stencil achieves EOC $\approx$ 2.94–2.98 for $N_x = 128$–256, demonstrating that the DPG cell-center values of ϑ_h carry enough superconvergence information to support third-order Gamma recovery. This breaks down at $N_x = 512$ due to temporal saturation; $N_t \geq 2048$ would be needed to observe the cubic rate at that resolution.

5. **L^2-Proj-P_2: good, moderately irregular.** Non-overlapping macro-element P_2 gives EOC $\approx$ 1.3–2.3, performing similarly to LSQ-P_2 without a sliding window.

6. **Temporal saturation at $N_x = 512$.** Fixing $N_t = 512$ makes the temporal error the dominant contribution at $N_x = 512$. All Greek errors worsen or stagnate. The Gamma study should use $N_t \geq 2048$ to isolate spatial convergence at this resolution.

7.2 Physical interpretation

The finding that LSQ reconstruction outperforms raw differentiation is consistent with the classical theory of superconvergent patch recovery (SPR) [1]: element-average (cell-center) values of DPG trial functions are superconvergent sampling points, carrying $\mathcal{O}(h^2)$ accuracy even though the trial space is only $\mathcal{O}(h)$ accurate in the interior of elements. Over-determined polynomial fitting over these superconvergent points extracts the full $\mathcal{O}(h^2)$ (or higher) information and delivers it as a smooth derivative estimate.

The within-element slope coefficient b, by contrast, does not benefit from this superconvergence: it is determined by the global DPG solve and can exhibit $\mathcal{O}(h)$ oscillations between adjacent elements.

7.3 Recommendations for future work

- Use $N_t \geq 2\,048$ for the Gamma convergence study to keep the finest mesh ($N_x = 512$) out of temporal saturation.
- For production Gamma output, use LSQ-P_2 (9-cell stencil): robust, monotone, $\approx O(h^2)$ convergence.
- LSQ-P_3 is preferred for highest accuracy on pre-saturation meshes ($N_x \leq 256$, $N_t = 512$).
- Consider extending the reconstruction to Vega and Rho, which also require differentiation of the DPG trial solution.

8 Files Produced

File	Description
`src/main_european_1d_ultraweak_mpi.cpp`	Fixed `gamma_exact`; new `--n2-csv` flag
`scripts/run_n2_gamma.py`	Runner: Singularity container, postprocessin
`scripts/plot_n2_gamma.py`	Figures N2a (convergence) and N2b (profiles
`scripts/generate_table_N2.py`	LaTeX table & paper text blurb
`results/greeks/v2_gamma_reconstruction.csv`	20 rows: 4 methods × 5 meshes
`results/greeks/v2_gamma_reconstruction_snapshots.csv`	59 rows: Gamma profiles at $N_x = 512$, $S \in [$
`results/figures/figN2a_....pdf/.png`	Log-log convergence plot (PDF 23 KB, PNG
`results/figures/figN2b_....pdf/.png`	Gamma profiles at $N_x = 512$ (PDF 23 KB, P
`results/paper_tables_siam.tex`	`\tableGammaReconstruction` appended
`results/paper_text_siam.tex`	`\textGammaRecovery` appended

A Acceptance Check

The following check was executed after the run:

```
--- Acceptance check ---
  Raw Gamma EOC (mean): 1.268
  LSQ-P2 Gamma EOC (mean): 1.637
  Phase N2 acceptance: PASS   (min LSQ-P2 EOC = 1.42 > 0.3)
```

B C++ Changes Detail

Bug fix (gamma_exact)

In `extract_greeks` (around line 440):

```
// Before (WRONG):
g.gamma_exact[i] = npdf(d1) / (g.S[i] * sigma * sq);

// After (correct):
g.gamma_exact[i] = npdf(d1) / (g.S[i] * sq);  // sq = sigma*sqrt(tau)
```

Within-element gradient (`vtheta_dx_intra`)

Added to the element loop in `extract_greeks`:

```
IntegrationPoint ipc_l, ipc_r;
ipc_l.Set2(0.25, 0.5);   // left quarter point
ipc_r.Set2(0.75, 0.5);   // right quarter point
const double sig_x_l = sig_gf.GetValue(i, ipc_l, 1);
const double sig_x_r = sig_gf.GetValue(i, ipc_r, 1);
const double vtheta_dx_intra = (sig_x_r - sig_x_l) / (diff * 0.5 * h);
```

Gathered via MPI and stored as `g.vtheta_dx_intra[i]`.

New `--n2-csv` output

Columns: `x_c, S_c, vartheta_c, vtheta_dx_raw, delta_exact, gamma_exact`.
Written after the final time step by rank 0 if the flag is supplied.

References

[1] O. C. Zienkiewicz and J. Z. Zhu, "The superconvergent patch recovery and a posteriori error estimates. Part 1: The recovery technique," *Internat. J. Numer. Methods Engrg.*, 33(7):1331–1364, 1992.